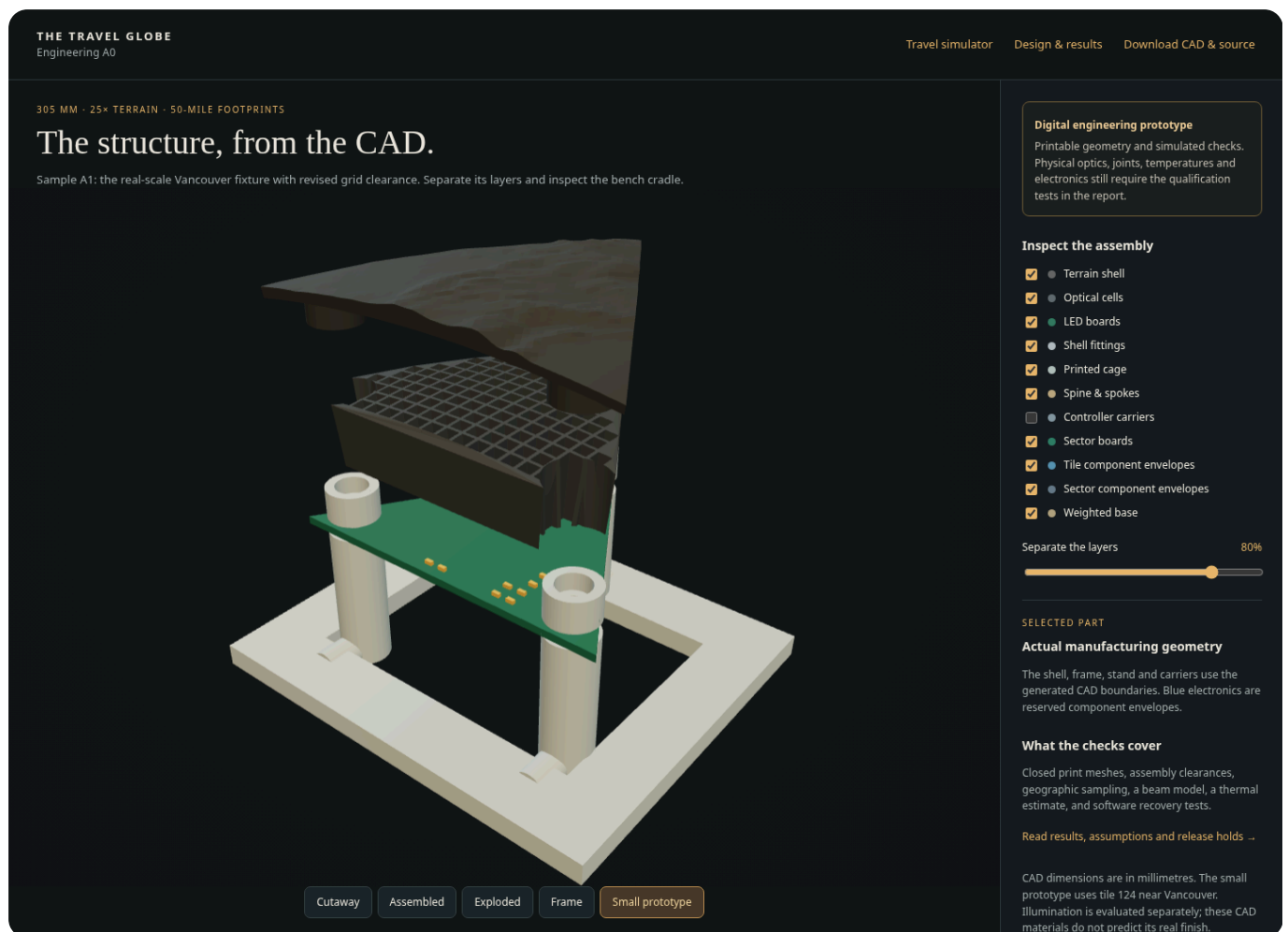


Manufacture the first globe sample

Build one interchangeable curved optical fixture at the final globe's scale. Keep the 305 mm diameter, 25× terrain and 50-mile patterns. This sample tests light transmission, unwanted glow, finish, print detail and fit. It does not establish complete-globe strength, power consumption, longevity or matrix-driver operation.

Print-shop request ZIP · PCB-assembly request ZIP · Test plan PDF · Inspect the fixture in 3D

Current sample revision: A1. This includes a revised grid, new finish/thickness coupons and checked bench firmware. Read the specific component/material choices and outstanding reviews. The packages are ready for supplier review, **not fabrication**. LED datasheet approval, printer confirmation and physical finish qualification remain open. An added solder-envelope check also found a potentially inadequate grid clearance: review assembled LED/solder dimensions and add relief if needed before ordering the grid. No supplier response has been received.



What to order

Use two service requests: one for the printed parts and one for the assembled circuit board. Ask each service to review its files and quote before fabrication. No supplier has been selected, contacted or paid, and these files have not yet been physically qualified.

Examples of services to request quotes from are PCBWay, or JLC3DP for printing and JLCPCB for assembly. These are candidates, not qualified suppliers for these particular files. Their review must confirm the thin walls, clear resin and chosen LED before you place an order.

Printed parts

The fixture footprint is approximately 56 × 48 mm. Everything is in millimetres at 100% scale. A1 enlarges only the sample grid's boss openings and adds a stepped material coupon. The full-globe A0 design is unchanged. Use these current request packs; the older `optical-prototype-A0.zip` does not contain these revisions.

File in the print request	Quantity	Requested material/process	Purpose
<code>prototype-shell-vancouver-25x.stl</code>	2	Request Formlabs Clear Resin V5; same batch	Keep one clear; tint the other
<code>prototype-shell-himalaya-25x.stl</code>	2	Same clear resin/process	Thicker-terrain comparison; one clear and one tinted
<code>sample-baffle-type-02-clearance.stl</code>	2	Request Formlabs Black Resin V5	Revised boss clearance; working grid and spare
<code>sample-finish-thickness-coupon.stl</code>	2	Same clear resin as the shells	One uncoated control; one finish trial at 1, 2, 3 and 4.75 mm thickness
<code>prototype-cradle-type-02.stl</code>	1	SLS/MJF PA12 preferred	Bench fixture; a rigid alternative can be quoted separately
<code>magnet-seat-M2-4mm.stl</code>	3	PA12, matched to the fit coupon	Magnetic shell seats
<code>magnet-fit-coupon.stl</code>	1	Same material/process as the seats	Check magnet-pocket fit before assembly

Ask the printer to confirm the **0.4 mm baffle walls**, approximately 1.6 mm cell openings, 1 mm minimum shell wall and magnet pockets. Layer height alone does not establish dimensional accuracy. Request the actual resin product, print process, orientation, post-cure process and expected dimensional tolerance. Ask about optical clarity, yellowing/UV stability and compatibility with a transparent tint and matte clear coat.

The named resins are first-choice quote requests, not confirmed products offered by these services. Ask for the exact alternative product if unavailable. The qualification sheet specifies the tolerance request and a supplier response form is inside each ZIP.

Do not permit automatic scaling, thickening, hollowing or geometry repair that changes these features. Ask the service to flag a feature it cannot print. Agree support placement before printing: protect the smooth inner optical surface, shell seating faces, magnet pockets and baffle openings. Deliver washed and fully cured parts, with every cell cleaned, unpainted and undyed. Keep post-processing identical between the clear controls and finish samples; avoid sanding or polishing the optical surfaces without recording it.

Black appearance alone does not prove a 0.4 mm wall blocks amber light. That is part of this sample's test. Do not silently substitute ordinary opaque white resin for the shell or a thicker coarse grid for the baffle.

Assembled optical PCB

Request the board service's minimum economical bare-board quantity, with **two boards assembled** if the quote is acceptable. One is the working board; the other helps distinguish an assembly fault from an optical problem. Do not buy a quantity of full-globe matrix boards.

The PCB request contains the routed `optical-bench-10` board, Gerbers, separate plated/nonplated drills, BOM, component placement CSV, project/library files and its saved-file validation. Specify **two layers, 1.0 mm FR4, nominal 35 µm copper, black soldermask preferred**. The ten 1005-metric/0402-imperial LEDs are on the front; ten 1 kohm 0603 resistors are on the back. Wire pads are not components to assemble.

For an automatic PCB quote, upload the enclosed `fabrication-gerbers.zip`. Upload `bom.csv` and `placement.csv` separately for assembly, and send `START-HERE-RFQ.txt` with the order-review request. For printing, upload each STL under its specified material and quantity from `print-order.csv`.

The BOM's Kingbright APHHS1005SYCK LED is a **candidate**, not a qualified final choice. Require the assembler to check the current manufacturer datasheet against the footprint, cathode/pad-1 assignment, package size, placement rotation and amber/yellow color. Ask them to identify any substitute for review before fitting it. Require continuity/short inspection and an individual LED test through the fitted resistors; an automated optical inspection alone does not prove correct light output. Keep the datasheet and inspection result with the sample.

The tiny LED/resistor assembly is outsourced. Your soldering is the eleven accessible wires between the finished board and a Pico.

Other items

Buy one Raspberry Pi Pico **with headers**, a data-capable USB cable, fine insulated wire and ordinary soldering supplies. Use a small screw-terminal/header adapter if it makes Pico wiring easier. Also obtain:

- Twenty 4 mm diameter × 2 mm magnets: fifteen are used for the three seats and four shells, leaving five spare.
- Three M2 × 22 countersunk screws, three M2 nuts, and four adhesive rubber feet at least 2 mm thick. Confirm the screw-head fit and engagement on the printed parts.
- A removable magnet-retention material compatible with the chosen resin; qualify it on the fit coupon. Keep the seat screws serviceable.
- A caliper, basic multimeter and contact thermometer if available.
- For the coupon experiment: **Tamiya X-19 Smoke**, optional **X-20A** acrylic thinner, and **XF-86 Flat Clear**. These are proposed products; resin compatibility and current application instructions still need review. Test on the new coupon before coating terrain, using the documented sequence.

Use **\$250 as the first-round target**, and a **\$500 stop-and-review ceiling** within the overall globe budget. These are suggested spending limits, not supplier prices. Quote printing, assembly, materials and shipping separately; manufacturing minimums may dominate the cost.

Assembly and first power

1. Photograph and label every part: Vancouver clear/tint, Himalaya clear/tint, baffle A/B. Record resin and batch. Measure the magnet coupon and inspect all optical cells under a light. Do not force a magnet or a warped shell into place.
2. With the circuit unpowered, inspect polarity, solder joints and shorts. Confirm each numbered wire pad reaches one LED anode through its 1 kohm resistor and that the cathodes share ground.
3. Solder board pads **0-9 to Pico GP0-GP9**, respectively, and board GND to Pico GND. GPIO names are not physical header-pin numbers; `firmware/wiring.csv` gives the exact header-pin mapping. Add wire strain relief at the cradle. This is a USB-powered logic-level circuit: **never apply the full globe's 12 V supply**.

4. Install Thonny and the **original RP2040 Pico** build of MicroPython. Select v1.26.0 where available to match the tested language runtime; a different build needs the same checks on hardware. Copy `firmware/main.py` and `firmware/patterns.json` from the PCB request to the Pico's root. On the first boot the LEDs are off; after a valid pattern is saved, it returns on restart. Invalid saved settings boot dark. No CAD software or KiCad is required to run the physical optical test.
5. In Thonny's Shell run `import main`, then `main.walk(0.1, 1000)`. Confirm ten separate emitters illuminate in numbered order for one second each, then go dark. For a steady single emitter use `main.show('walk-0', 0.1)` through `walk-9`. The resistors stay fitted for every test.
6. Install the nuts, board and three magnet seats in the cradle using the M2 × 22 screws. Fit the rubber feet. Fit the baffle without stressing the board or blocking cells; small removable retention dots at the outer edge are acceptable. Keep adhesive out of the light paths.
7. Pair and mark magnet polarities. Each shell uses three magnets and shares the fixture's three mating magnets. The seating rim at local Z=5.0 mm sets shell position; the fixture magnet face is recessed at Z=4.8 mm. The baffle must not support or push the shell outward. Test retention and removability gently.
8. Test **both clear shells before applying any finish**. If the clear shell is already too dim or creates a broad haze, investigate that before adding tint.

A controlled finish experiment

Keep one shell of each terrain uncoated throughout. On the other pair, test a thin transparent tint first; photograph and record it after full cure. Add another coat only if necessary, recording the change. Apply matte clear only after evaluating the tint itself, then repeat the comparison. This separates the effects of resin thickness, pigment and surface finish.

First qualify the coating on the two stepped witness coupons using the specific finish experiment. Keep the underside uncoated. Record the resulting recipe rather than assuming the named paint/resin combination works.

Use the same PCB, baffle, LED pattern, viewing distance, room lighting and PWM setting for paired comparisons. Do not change the 50-mile pattern or enlarge the illuminated patch to make a poor material result look better. If you decide to enlarge footprints, record it as a deliberate design tradeoff.

Try these commands:

```
main.show('walk-0', 0.1) # one emitter; repeat walk-1 through walk-9
main.show('vancouver', 0.1)
main.show('seattle', 0.1)
main.show('both', 0.1)
main.show('all', 0.1)
main.show('off', 0)
```

`show()` saves by default. Pass `False` as a third argument for temporary measurements, e.g. `main.show('all', 0.1, False)`, to avoid unnecessary flash writes. `walk()` never saves and ends dark; it leaves the last saved pattern unchanged. After normal shutdown, unplug/replug USB and verify the last deliberately saved pattern returns. A failed write blanks outputs and reports an error; report and resolve it before continuing. Host runtime tests do not establish power-loss durability of Pico flash.

Repeat the comparisons at PWM scales 0.1, 0.3 and 1.0 as needed; **1.0 means full PWM, not 1 amp or a calibrated brightness**. Keep the 1 kohm resistors in place. For an individual LED at full PWM, measuring voltage across its resistor gives current approximately as $I = V / 1000 \text{ ohms}$. Record the result rather than assuming the bench matches the eventual scanned matrix driver.

Photograph unlit and lit samples in normal room lighting, then in dim conditions. Check the object by eye at approximately 25 cm and 1 m. Lock camera exposure, focus and white balance; turn off HDR/night-mode processing and avoid saturated highlights. An ordinary phone JPEG is useful for visual comparison but **not a calibrated light measurement**. Use RAW/linear image data or suitable optical instrumentation for quantitative brightness ratios, and subtract the LEDs-off background.

What would constitute a useful pass?

These are proposed acceptance targets, not measured results or guarantees.

Use `sample-results.csv` in either request pack to record results, conditions and photo filenames. Leave a failed or unperformed test visibly marked; do not fill it with a simulated result.

Test	Evidence to record	Starting decision rule
Finish when off	Clear/tinted pair under normal room light	Tinted shell looks acceptably charcoal while preserving visible terrain
Local light	One-cell walk and the 50-mile city patterns	Warm, localized patches; no broad glow through the whole shell
Neighbor leakage	Single commanded cell with off-frame background removed	Target brightest adjacent uncommanded cell below 15% of the lit cell, using a linear measurement
Thick terrain	Swap Vancouver/Himalaya shells on the same board, comparing the same emitter at unchanged settings	Target at least 40% of the reference-region light; investigate dim ridges before considering compensation
Two destinations	Vancouver, Seattle, then both	Their relative positions and separation remain understandable; use a geographic reference for context
Fit and service	Photograph seated gaps; remove/refit at least 20 times	No flexing, sticking, cracked walls or migrating magnets; shell seats consistently
Stability of the sample	Run selected pattern for one hour; record current and contact temperature	No visible finish change, loosening or unexpected heating; this does not qualify the full globe's thermal limits

The small triangular sample and its ten selected emitters cannot establish whole-globe geographic recognizability. Assess the size of the local marks now, then use a larger populated panel for that decision. The high-relief coupon samples thicker terrain; it does not necessarily put the globe's maximum 4.73 mm thickness over every tested emitter.

If the optical result only works in a dark room at the bench board's highest output, treat it as marginal until a final-driver test establishes sufficient light within the power budget. If tint blocks mountains or clear resin spreads too much light, change the material/finish or shell construction and repeat this inexpensive sample before ordering a full globe.

The next sample after this passes

The next release should combine **one completed final-driver LED tile** with the winning shell/baffle material, plus **one full-size terrain panel and representative cage/spoke joints**. The matrix tile still needs completed routing and electrical review before it can be manufactured. That stage checks real scanned-drive brightness, current and

shutdown; panel warpage and seams; retention; and joint fit/load behavior. A later populated sector checks geographic recognition, cabling and heat.

Keep the successful material/process recipe, uncoated controls, photographs, current readings and test log. Passing the small optical fixture is a decision to proceed to those tests, not approval to order all 226 matrix boards or all twenty terrain panels.

Sample A1 digital review. No supplier order or physical test has been performed.

Sample A1: component and material review

Status: prepared for supplier review and a physical experiment; not released for fabrication. The LED manufacturer drawing, current stock, printer capability and finish compatibility are still open. There is also a solder-to-grid clearance hold described below. No supplier has confirmed these items. This document records the exact proposed choices and the evidence needed to close each item.

Sample instructions · Printer request · PCB request

Proposed materials and finish

Part	First-choice quote request	What must be confirmed
Four curved shells and two thickness coupons	Formlabs Clear Resin V5 , untinted, fully washed/cured, same resin lot and post-processing	Printer actually uses this product; 1 mm minimum shell wall; support strategy; cure; dimensional tolerance; clarity and UV/yellowing information
Two cell grids	Formlabs Black Resin V5 , fully washed/cured	Unsupported 0.4 mm walls and approximately 1.6 mm open cells survive printing and cleaning; optical opacity is a subsequent sample test
Cradle, three seats, magnet fit coupon	PA12 , SLS or MJF; use the same process for seats and fit coupon	State powder product/process and dimensional tolerance; no paint or infiltration that changes fits
Transparent darkening experiment	Tamiya X-19 Smoke acrylic model paint	Current product instructions and supplier advice for the particular fully cured resin; physical coupon adhesion/transmission test
Thinner, only if needed	Tamiya X-20A acrylic thinner	Use only the product's stated method; record actual dilution; no improvised solvent substitution
Separate matte topcoat experiment	Tamiya XF-86 Flat Clear acrylic	Qualify over cured X-19 on the witness coupon before a terrain shell

These are specific **experimental candidates**, not verified resin/paint combinations or claims that PCBWay/JLC3DP stock them. If the service cannot supply the named resin, request its exact alternative product and technical data sheet as a separate option. Keep control and coated parts on the same alternative resin and batch. Do not buy a litre of resin for this outsourced test.

Product identities, supplier availability and current application instructions still need primary-source review. The websites to use are Formlabs, Formlabs support, and Tamiya USA. They are reference destinations, not evidence of a completed review.

Finish trial before coating terrain

The new 32 × 24 mm coupon has four 8 mm strips, **1, 2, 3 and 4.75 mm thick**, with a flat underside. Order two in the same clear resin batch as the shells. Keep coupon C0 completely uncoated. On coupon C1, divide its 24 mm length into three 8 mm lanes running across all four thicknesses; label only its edge or a separate card.

1. Photograph both coupons unlit and backlit before coating. Reject oily, tacky, cracked or visibly under-cured surfaces to the printer. Use the coating maker's preparation and curing instructions; do not invent an abrasive or solvent treatment.
2. Brush one thin, even X-19 coat on C1, leaving its underside clear. Record brush, dilution, coat count, temperature and cure time. After the stated cure, photograph it at the same LED setting and exposure as C0.
3. Add one further thin X-19 coat to lanes B and C only; cure and compare. Then apply one thin XF-86 coat to lane C only, following its instructions. The three lanes now compare one smoke coat, two smoke coats, and two smoke coats plus matte clear.
4. Place the same coupon strip over the same light source at unchanged distance for each comparison. Use an opaque card mask to isolate one strip at a time; this is a material transmission comparison, not the curved fixture's leakage test.
5. After the stated cure and again after seven days, inspect for tack, clouding, cracks, delamination and handling marks. On the coupon edge only, try a gentle thumbnail rub and a repeatable low-tack tape lift. Record tape and method. This is a practical screening test, not an ASTM adhesion certification or a longevity prediction.
6. Choose the least coating that looks acceptably charcoal when off while leaving the thick strip visibly lit. Transfer that recorded recipe to one Vancouver and one Himalaya shell, exterior only. Keep the two other shells clear. Repeat the curved fixture tests before accepting the recipe.

If no lane passes both appearance and transmission, stop and change the resin/finish; a thicker opaque paint coat cannot solve that tradeoff. The witness coupon checks the worst nominal wall thickness independently of where the ten sample LEDs happen to sit under mountains.

LED and resistor review

The first candidate remains **Kingbright APHHS1005SYCK**, ten front-side LEDs per board. Request its current manufacturer PDF from Kingbright USA, plus distributor stock/lead time from DigiKey or Mouser. A catalog description saying "0402" is insufficient.

The saved board is electrically routed, but its LED footprint is a **generic KiCad 1005-metric LED footprint**, not proof of fit to that Kingbright part. These facts can be checked from the supplied board:

Board fact	Value to compare against the manufacturer drawing
Fabrication body rectangle	1.0 × 0.5 mm
Fit-audit body allowance	1.0 × 0.5 × 0.5 mm; assumed, not a verified package maximum
Copper pad size	0.59 × 0.64 mm each
Pad centers	X = −0.485 and +0.485 mm in footprint coordinates
Copper inner gap / total span	0.38 / 1.56 mm
Electrical convention	Pad 1 = cathode/common GND; pad 2 = anode through its channel resistor
Placement	D1–D10 on front, rotations in <code>placement.csv</code> ; manufacturer cathode mark and tape orientation must be reconciled with these rotations

The assembler must return the package maximum dimensions, recommended land pattern, cathode marking/drawing, emission direction, wavelength/color bin, current rating, forward-voltage range, reflow profile and an in-stock sourcing option. Record PDF revision and distributor/date in `supplier-review.csv`. Do not populate an unreviewed substitute. If the land pattern differs, revise and recheck the board before ordering.

Solder clearance hold: the grid starts 0.10 mm above the nominal PCB face. An additional check extruded each entire LED copper pad to an **assumed 0.20 mm solder height**; the smallest side clearance to the grid was just **0.0127 mm**. That allowance is not a measured solder fillet, but it shows why the much larger LED-body clearance cannot establish manufacturability. Obtain the actual package underside/terminal drawing, assembled height, paste/fillet envelope and board/grid positioning tolerances. If solder occupies this allowance, add local underside relief to the grid and recheck the geometry and optical leakage before printing. Do not rely on a printer holding a 0.013 mm side gap or force the grid down onto solder joints. **The current sample grid remains on hold for this review/redesign.**

The resistor candidate is **Yageo RC0603FR-071KL**, 1 kohm, 1%, 0603 imperial, ten rear-side parts. Confirm its current data sheet and minimum 0.063 W rating or approve an equivalent. At an **assumed** maximum logic output of 3.465 V and minimum 990 ohms, even a shorted LED would draw at most about **3.50 mA per channel**, 35 mA for ten, and dissipate about **12.2 mW in each resistor**. This conservative resistor calculation does not verify Pico per-pin/aggregate output ratings, actual LED current, brightness, or the final matrix driver's operating point. Check the original RP2040 Pico documentation as part of the electrical review.

Printer reply needed

Use PCBWay 3D printing or JLC3DP for a capability/quote request. Use PCBWay assembly or JLCPCB for the passive board. These are candidate services; no account submission, quote, availability check or order has occurred.

Ask for a written reply to all rows of `supplier-review.csv`, including exact material products, support placement, cleaning, cure and dimensional tolerances. **Request ± 0.10 mm or better on the mating optical faces as a proposed target, not an established process capability.** The revised grid's nominal minimum shell gap is 0.349 mm. Two surfaces each displaced 0.10 mm toward each other would leave about 0.149 mm; warpage, accumulated mounting errors and resin swelling are additional. If the service cannot meet the target, return its tolerance estimate for a clearance redesign instead of forcing the fit.

Boss apertures are now 7.6 mm around nominal 6.6 mm bosses. Verify they remain open after support removal. Preserve all 0.4 mm walls and clean every cell. Magnet fit must still be checked against the supplied pocket coupon and actual magnets; nominal magnet diameter is not a press-fit specification.

Evidence already obtained

- New baffle and stepped coupon: saved STLs each form one closed, consistently oriented solid with positive volume.
- Both curved shells: no unexpected surface intersections with the nominal assembly; independent solid intersections show no baffle overlap with other parts. Minimum modeled shell/grid gap is **0.349 mm**; the ten assumed LED bodies have at least **0.270 mm** to the grid.
- Added solder-envelope check: minimum **0.0127 mm** side gap for an assumed 0.20 mm-high envelope over each full LED copper pad. This is a **fabrication hold**, not a passed tolerance check; actual assembled dimensions are required.
- Existing passive PCB: zero saved DRC violations and zero unconnected nets; routing unchanged.

- Revised firmware: **11 behavioral cases passed in CPython and MicroPython v1.26.0 Unix**, including all ten channels, invalid pattern rejection, damaged saved settings, interrupted writes, GPIO/storage errors and an interrupted walk. GPIO and timing were mocked; the filesystem was the host's. This is not a Pico execution or flash power-cut test.

The JSON evidence is supplied in both request packs. Pending items are deliberately left blank in the supplier review sheet. A written printer reply, approved LED data sheet and material coupon results are required before describing the corresponding choices as qualified.

Sample A1 digital review. No supplier order or physical test has been performed.